

DE Lab12
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Q1:

first going to label all the transistors

left inverter N1 P1

right inverter N2 P2

left to BL WL1

right to notBL WL2

PC BL PC1

PC notBL PC2

PC BL to notBL PC3

i)

i.i)

assume $Q=0$, $\text{not}Q=VDD$

circuit already precharged $BL=\text{not}BL=VDD$

$VGS(N1)=VDD$, $VSG(P1)=0$

stated that WL fully ON

N1 ON

P1 OFF

N2 OFF

P2 ON

WL1 ON

WL2 ON

PC1 OFF

PC2 OFF

PC3 OFF

i.ii)

N1: linear

$G = \text{not } Q \sim = VDD$

$S = 0$

$D = Q \sim = 0$

$VGS = VDD - 0 = VDD$

$VDS = Q - 0 = 0$

saturation requires $VDS \geq VGS - V_T$

$VDS? \geq VGS - V_T$

$0? \geq 1.2 - 0.4 = 0.8$, fails, linear

P2: linear

$G = Q \sim = 0$

$S = VDD$

$D = \text{not } Q \sim = VDD$

$VSG = VDD - 0 = VDD$

$VSD = VDD - VDD = 0$

saturation requires $VSD \geq VSG - |V_T|$

$VSD? \geq VSG - |V_T|$

$0? \geq 1.2 - 0.4 = 0.8$, fails, linear

WL1: saturation

$G = WL = VDD$

$S = Q \sim = 0$

$D = BL = VDD$

$VGS = VDD - 0 = VDD$

$VDS = VDD - 0 = VDD$

saturation requires $VDS \geq VGS - V_T$

$VDS? \geq VGS - V_T$

$1.2? \geq 1.2 - 0.4 = 0.8$, succeeds, saturation

WL2: linear

$G = WL = VDD$

$S = \text{not } Q \sim = VDD$

$D = \text{not } BL = VDD$

$VGS = VDD - VDD = 0$

$VDS = VDD - VDD = 0$

saturation requires $VDS \geq VGS - V_T$

$VDS? \geq VGS - V_T$

$0? \geq 0 - 0.4 = -0.4$, fails, linear

P1, N2, PC1, PC2, PC3: off

i.iii)

Q: 0.338V

WL1 pulling up to BL=VDD

N1 pulling down to 0

solve $I(WL1)=I(N1)$

$I(WL1) = \text{saturation} = \frac{1}{2} k_n W/L (V_{GS}-V_T)^2 = \frac{1}{2} k_n 2 ((1.2-Q)-0.4)^2 = k_n (0.64-1.6Q+Q^2)$

$I(N1) = \text{linear} = k_n ((V_{GS}-V_T)V_{DS} - V_{DS}^2/2) = k_n ((1.2-0.4)Q - Q^2/2) = k_n (0.8Q - 0.5 Q^2)$

$0.64-1.6Q+Q^2 = 0.8Q - 0.5 Q^2$

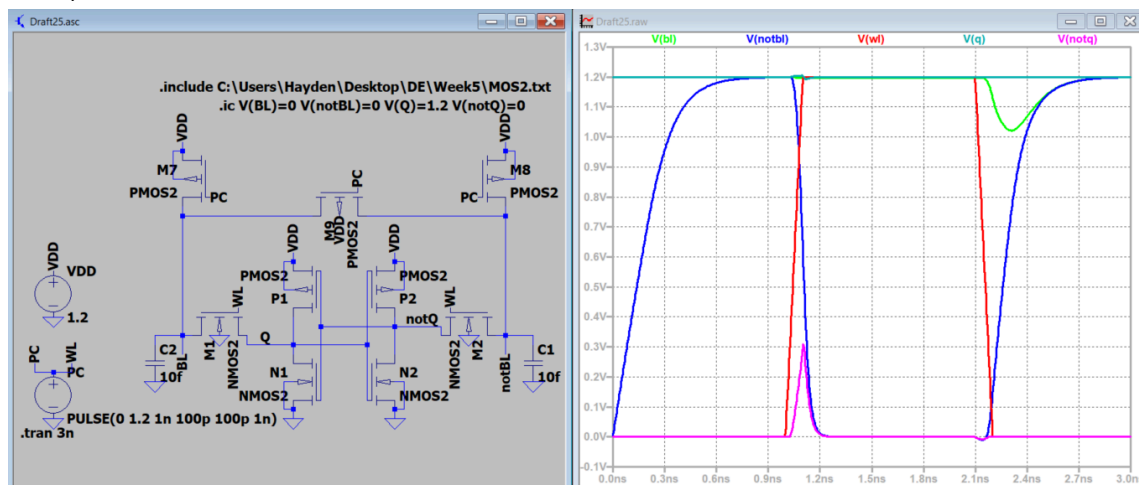
$0.64-2.4Q+1.5Q^2=0$

$Q=0.338119784648V$

notQ: ~1.2V

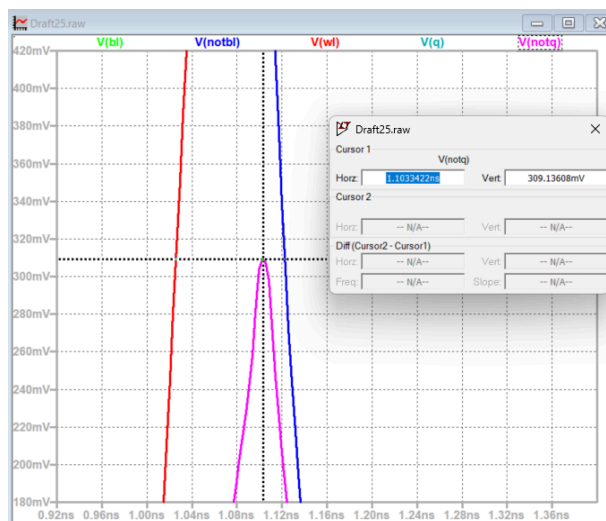
pulled up to VDD by both P2 and WL2, so notQ~VDD

ii + iii)



iv)

0.309V



Q2:

a)

$$\Delta_{BL} = VDD/2 * CB / (CB + n * CC) = 0.6 * 25 / (25 + n)$$

$$\Delta_{BL} \geq 0.1V$$

$$0.6 * 25 / (25 + n) \geq 0.1$$

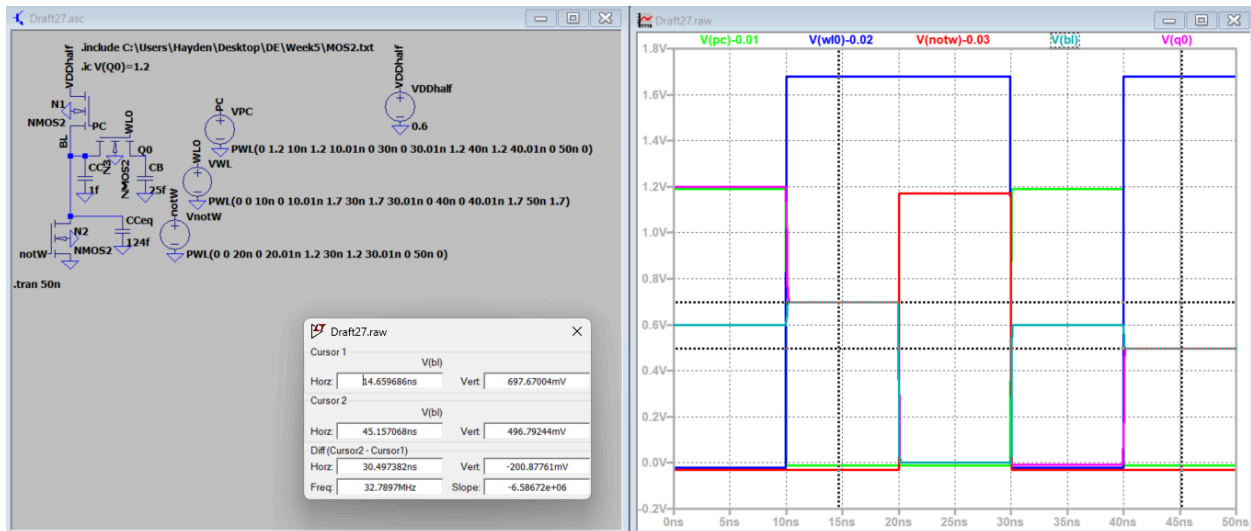
$$150 \geq 25 + n$$

$$n \leq 125$$

$$n = 125$$

b)

precharge, read 1, write 0, precharge, read 0



we found at while reading BL = 697.67mV and 496.79 for 1 and 0 respectively. These are both slightly lower than expected, pulling the 1 slightly lower than the required 700mV threshold. I found that in our simulation we reach the 700mV threshold with n=121.